

Bridging the Tokenization Gap in African LLMs via SuperBPE

Team: Bangweulu

Programme: TRI AI Saturdays — Cohort 10

Problem statement

Large language models often struggle with African languages because current tokenizers are English-centric, leading to inefficient processing, high computational costs, and poor linguistic representation. This issue significantly impacts African AI developers, translators, and millions of speakers, limiting access to effective digital tools. Our project addresses this by developing a superBPE tokenizer trained on diverse African language corpora to improve semantic accuracy and efficiency. Guided by the values of inclusivity, fairness, cultural preservation, transparency, and Ubuntu, our solution aims to reduce inference costs while preserving linguistic diversity. By providing a high-quality, reusable tool, we hope to enable the development of accessible, robust African language models, empowering local communities and fostering equitable participation in the global AI ecosystem.